Chapter 4 - Network Layer

Intro

Forwarding and Routing
Network Service models
Virtual Circuit and Datagram Networks
Inside a Router
IP (ipv4 & ipv6)
Routing Algorithms

Maximum Host IP:

The maximum usable host IP is one less than the broadcast address:

- **Broadcast Address**: 10.35.15.255
- **Maximum Host IP**: 10.35.15.254

Private IP addresses are defined by three main ranges reserved by the Internet Engineering Task Force (IETF) and are not routable on the public Internet. These addresses are used for local communication within private networks (e.g., home, office LANs). The ranges are:

1. **Class A (10.0.0.0 to 10.255.255.255 - 10/8)**: This range provides 16,777,216 addresses.
2. **Class B (172.16.0.0 to 172.31.255.255 - 172.16/12)**: This range provides 1,048,576 addresses.
3. **Class C (192.168.0.0 to 192.168.255.255 - 192.168/16)**: This range provides 65,536 addresses.

Routing Algorithms

Comparison of LS and DV algorithms

message complexity

LS: n routers, $O(n^2)$ messages sent

DV: exchange between neighbors; convergence time varies

speed of convergence

LS: $O(n^2)$ algorithm, $O(n^2)$ messages
- may have oscillations

DV: convergence time varies
- may have routing loops
- count-to-infinity problem

robustness: what happens if router malfunctions, or is compromised?

LS:
- router can advertise incorrect *link* cost
- each router computes only its *own* table

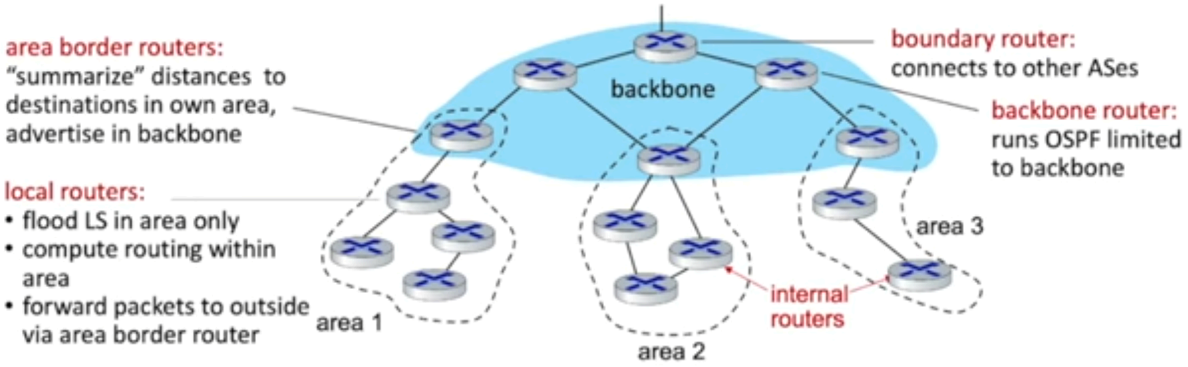
DV:
- DV router can advertise incorrect *path* cost ("I have a *really* low-cost path to everywhere"): *black-holing*
- each router's DV is used by others: error propagate thru network

Intra Autonomous routing Protocols

1. RIP : Routing Information Protocol : Classic Distance Vector Algo.
    1. OSPF : Open Shortest Path First : Classic Link State Routing
2. EIGRP : Enhanced Interior Gateway Routing Protocol : DV Based

Hierarchical OSPF

- two-level hierarchy: local area, backbone.
  - link-state advertisements flooded only in area, or backbone
  - each node has detailed area topology; only knows direction to reach other destinations



Inter ISP Routing Algorithms

BGP: Border Gateway Protocol

Forwarding vs. Routing

Forwarding and **routing** are key concepts in the network layer, responsible for delivering data packets from a source to a destination.

Forwarding

- **What it is:** The process of moving a packet from one network interface to another within a router or switch.
- **Focus:** Happens at an individual device level (e.g., a router).
- **How it works:** Based on the **routing table**, the router decides which outgoing link/interface to use for the packet.
- **Example:** A router receives a packet and forwards it to the next router in the path to the destination.

Routing

- **What it is:** The process of finding the best path for data packets to travel from source to destination.
- **Focus:** Happens at a network-wide level.
- **How it works:** Routers exchange information and use algorithms (e.g., Dijkstra's or Bellman-Ford) to build and update **routing tables**.
- **Example:** A router updates its routing table to reflect the shortest path to a destination after receiving updates from neighboring routers.

Key Differences

Aspect	Forwarding	Routing
Scope	Single router's action	Whole network's path-finding
Goal	Move packet to the correct output	Determine the best path
Time	Happens in real-time for every packet	Occurs periodically or as needed
Type	Local process	Global process

**Network Service Models

Definition

The **network service model** describes the characteristics and guarantees provided by the network layer for end-to-end delivery of packets between sending and receiving hosts.

Types of Services Provided

1. **Guaranteed Delivery**
   - Ensures the packet will eventually reach its destination.
2. **Guaranteed Delivery with Bounded Delay**
   - Guarantees packet delivery within a specified time limit.
3. **In-Order Packet Delivery**
   - Packets arrive at the destination in the same order as sent.
4. **Guaranteed Minimum Bandwidth**
   - Ensures a minimum transmission rate (e.g., 1 Mbps) is available for the flow.
5. **Guaranteed Maximum Jitter**
   - Controls the variation in delay between packets for smoother data flow.
6. **Security Services**
   - Provides confidentiality, integrity, and authentication for the data.

Examples of Service Models

1. **Internet's Best-Effort Service**
   - No guarantees (packets may be delayed, dropped, or reordered).
   - Simple and scalable.
2. **ATM CBR (Constant Bit Rate)**
   - Guarantees fixed bandwidth with low delay (used for real-time traffic like voice/video).
3. **ATM ABR (Available Bit Rate)**
   - Guarantees a minimum bandwidth but allows higher rates if resources are free.
   - Provides congestion feedback to the sender.

Comparison Table (From Book)

Network	Service Model	Bandwidth	Loss Guarantee	Order	Timing	Congestion Info
Internet	Best Effort	None	None	Any order	Not maintained	None
ATM CBR	Constant Bit Rate	Guaranteed	Yes	In order	Maintained	Congestion won't occur
ATM ABR	Available Bit Rate	Guaranteed min	None	In order	Not maintained	Congestion feedback

Virtual Circuit and Datagram Networks

1. Virtual Circuit Networks

- **Connection-Oriented:** Establishes a pre-defined path between sender and receiver before sending data.
- **Setup Phase:** Requires a connection setup phase to reserve resources like bandwidth on the path.
- **Maintains State:** Routers maintain connection-specific information (e.g., path and reserved resources).
- **Packet Handling:** Packets follow the same path and arrive in order.
- **Examples:**
  - **ATM (Asynchronous Transfer Mode)**
  - **Frame Relay**
  - **MPLS (Multiprotocol Label Switching)**

Key Features:

- Ensures **reliable and ordered delivery**.
- Resources (e.g., bandwidth) are **reserved** for the connection.
- Efficient for applications requiring **consistent performance** like video conferencing.

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2. Datagram Networks

- **Connectionless:** Each packet (datagram) is treated independently.
- **No Setup Phase:** No need to establish a pre-defined path.
- **No State Maintenance:** Routers do not maintain information about packet flows.
- **Packet Handling:**
  - Packets may take different paths to reach the destination.
  - Packets can arrive out of order.
- **Examples:**
  - **Internet using IP (Internet Protocol)**

Key Features:

- Flexible and scalable.
- Suitable for bursty and unpredictable traffic.
- Relies on **best-effort delivery** without guarantees.

Comparison Table

Feature	Virtual Circuit Networks	Datagram Networks
Connection Type	Connection-oriented	Connectionless
Setup Phase	Required	Not required
State Maintenance	Yes, at routers	No
Packet Delivery Order	Guaranteed, ordered	Not guaranteed, unordered
Reliability Guarantees	High (resources reserved)	Best-effort
Flexibility	Low (fixed path)	High (dynamic path)

Use Cases

1. **Virtual Circuit Networks:**
   - Real-time applications like voice/video calls.
   - Applications requiring predictable delays.
2. **Datagram Networks:**
   - Web browsing, email, and file downloads.
   - Scenarios with unpredictable or bursty traffic.

What's Inside a Router?

1. Input Ports

- **Functions:**
  - Terminates the physical link at the router.
  - Performs **link-layer processing** (e.g., Ethernet, Wi-Fi).
  - **Lookup Function:** Determines the output port for an incoming packet by consulting the forwarding table.
  - Forwards control packets (e.g., routing protocol packets) to the routing processor.
- **Details:**
  - Input ports support both data forwarding and control-plane activities.
  - They can handle multiple ports on the same line card (common in ISP routers).

2. Switching Fabric

- **Definition:** Connects input ports to output ports within the router.
- **Purpose:** Transfers packets from input to output ports efficiently.

- **Analogy:** Acts as the "roundabout" in traffic where packets (cars) take paths to their destinations.

3. Output Ports

- **Functions:**
  - Buffers and stores packets coming from the switching fabric.
  - Performs **link-layer processing** and transmits packets onto the outgoing link.
- **Pairing:** Typically paired with input ports on the same physical link for bidirectional traffic.

4. Routing Processor

- **Role:** Handles control-plane functions, including:
  - Running **routing protocols** (e.g., OSPF, BGP).
  - Managing **routing tables** and computing **forwarding tables**.
  - Communicating with SDN controllers in modern networks for dynamic table updates.
  - Performing network management tasks.
- **Time Scale:**
  - Operates on slower time scales (milliseconds or seconds) compared to the data plane.

Key Considerations

- Most of the data plane (input/output ports, switching fabric) operates in **hardware** for speed (nanosecond time scales).
- Control plane functions are usually implemented in **software** on the routing processor.

Queuing in Routers

Definition

Queuing in routers refers to the process of temporarily storing packets in buffers when the outgoing link is busy. It occurs at both input and output ports and helps manage congestion and ensure efficient packet transmission.

Types of Queuing Delays

1. **Input Queuing:**
   - Occurs at input ports when packets cannot immediately pass through the switching fabric.

- Can lead to **Head-of-Line (HOL) Blocking**, where a packet at the front of the queue blocks others behind it, even if they are destined for free output ports.

2. **Output Queuing:**
   - Happens at output ports when packets wait for the transmission link to become free.
   - Congestion increases the queuing delay, leading to potential packet drops if the buffer is full.

Queuing Disciplines

1. **FIFO (First-In-First-Out):**
   - Packets are transmitted in the order they arrive.
   - Simple but does not prioritize packets based on importance.
2. **Priority Queuing:**
   - Packets are classified into priority levels (e.g., high or low priority).
   - High-priority packets are always transmitted first.
3. **Round-Robin Queuing:**
   - Alternates transmission between different classes of packets in a circular manner.
   - Ensures fairness among classes.
4. **Weighted Fair Queuing (WFQ):**
   - Assigns weights to classes; higher weights get more bandwidth.
   - Useful for guaranteeing minimum throughput for specific traffic classes.

Factors Affecting Queuing Delay

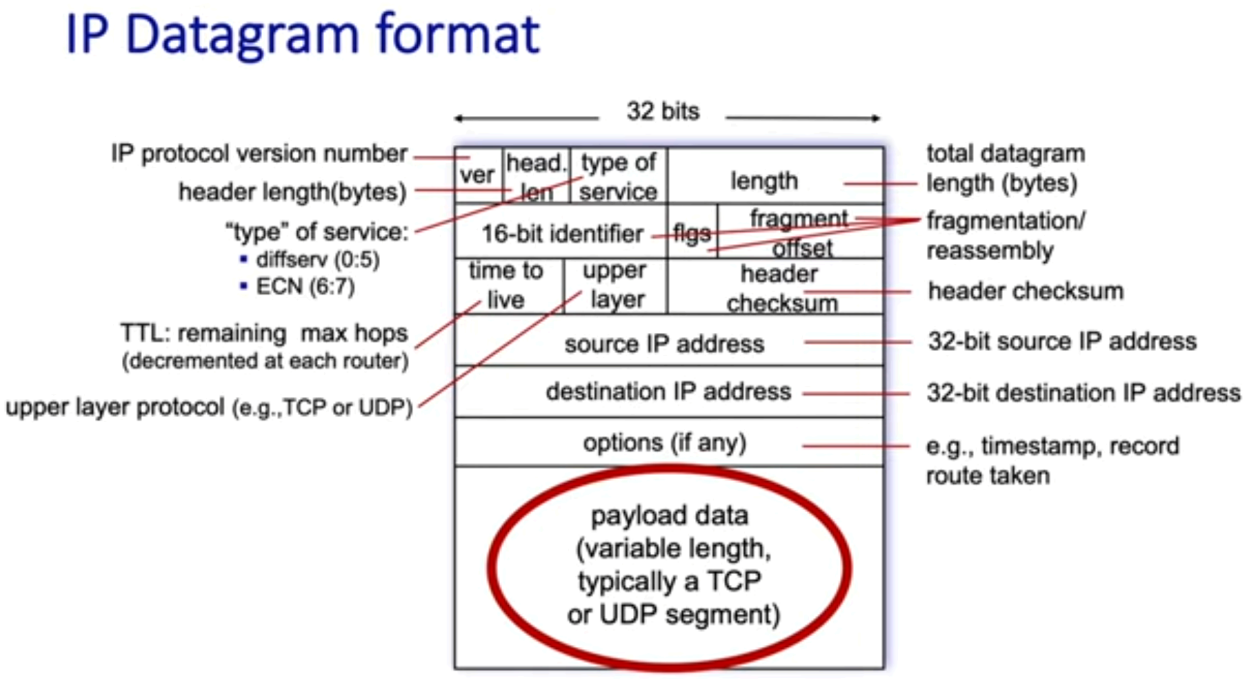
1. **Traffic Intensity (La/R):**
   - If the arrival rate of packets (La) exceeds the transmission rate (R), queues grow indefinitely, causing high delays or packet loss.
2. **Traffic Patterns:**
   - Bursty traffic leads to variable queuing delays.
   - Periodic or evenly spaced traffic reduces delays.

Consequences of Queuing

1. **Packet Loss:**
   - Occurs when buffers are full, and new packets are dropped.
   - Can be mitigated using Active Queue Management (e.g., RED - Random Early Detection).
2. **Increased Latency:**
   - High traffic can significantly increase queuing delays, especially as traffic intensity approaches

IP : Internet Protocol

IP Datagram Format Features



- **Fields**:
  1. **Version (4 bits)**:
     - Identifies the IP version (e.g., IPv4 or IPv6).
  2. **Header Length (4 bits)**:
     - Indicates the length of the IP header in 32-bit words.
  3. **Type of Service (8 bits)**:
     - Specifies the quality of service (e.g., low delay, high throughput).
  4. **Total Length (16 bits)**:
     - Specifies the total size of the datagram, including header and data.
  5. **Identification, Flags, Fragment Offset**:
     - Used for fragmentation and reassembly.
  6. **TTL (8 bits)**:
     - Limits the datagram's lifetime to avoid loops.
  7. **Protocol (8 bits)**:
     - Identifies the transport-layer protocol (e.g., TCP = 6, UDP = 17).
  8. **Header Checksum (16 bits)**:
     - Ensures header integrity.
  9. **Source and Destination IP (32 bits each)**:
     - Addressing for datagram routing.
  10. **Options (variable)**:

    - Optional settings for special purposes.
11. **Payload**:
    - The actual data being transmitted.

Classful Addressing

- Early IP addressing divided networks into classes:
  - **Class A**: `0.0.0.0` to `127.255.255.255` (large networks, few hosts).
  - **Class B**: `128.0.0.0` to `191.255.255.255` (medium networks).
  - **Class C**: `192.0.0.0` to `223.255.255.255` (small networks).
  - **Class D**: `224.0.0.0` to `229.255.255.255`
  - **Class E**: `240.0.0.0` to `255.255.255.255`

Longest Prefix Matching:

- Routers use **longest prefix matching** to determine the most specific route.
- Example:
  - Table contains:
    - `223.1.0.0/16` (general match).
    - `223.1.1.0/24` (more specific match).
  - Packet for `223.1.1.4` matches the longer prefix `/24`.

IP Fragmentation and Reassembly

- **Fragmentation**:
  - Necessary when a datagram exceeds the Maximum Transmission Unit (MTU) of a link (e.g., original datagram: 4,000 bytes, MTU: 1,500 bytes).
  - The datagram is divided into smaller fragments that fit within the MTU.
  - **Fragmentation fields in the IP header**:
    - **Identification**: Identifies fragments belonging to the same datagram.
    - **Flags**:
      - `1`: More fragments.
      - `0`: Last fragment.
    - **Fragment Offset**: Specifies the position of the fragment's data in the original datagram (measured in 8-byte chunks).
- **Reassembly**:
  - Done only at the destination to reconstruct the original datagram.
  - If one fragment is missing, the entire datagram is discarded.

- sending host increments the identification number for each datagram it sends , so fragments contain the identification number of the original datagram
- Examples:
  - **1st Fragment**: Offset = 0, 1,480 bytes of data.
  - **2nd Fragment**: Offset = 185 (starting at byte 1,480), 1,480 bytes of data.
  - **3rd Fragment**: Offset = 370 (starting at byte 2,960), 1,020 bytes of data.
- **IPv4 Vulnerabilities**:
  - Exploits like overlapping fragments or streams of small fragments can cause issues (e.g., Jolt2 attack).

IP Addressing and Subnetting

- **IPv4 Addressing**:
  - IP addresses are associated with **interfaces**, not hosts or routers.
  - Each IP address is globally unique (except for private networks).
  - Written in **dotted-decimal notation** (e.g., 223.1.1.4).
- **Subnets**:
  - A **subnet** is a logical subdivision of an IP network.
  - Identified by a subnet mask (e.g., `/24` for 24 network bits).
  - can be a collection of hosts and a router or between router interfaces.
  - Example:
    - Subnet `223.1.1.0/24`: Contains host addresses `223.1.1.1` to `223.1.1.254`.
    - Routers interconnect subnets.
- **CIDR (Classless Inter-Domain Routing)**:
  - Allows hierarchical allocation of IP addresses.
  - Written as `a.b.c.d/x`, where `x` specifies the prefix length.
  - Reduces the size of routing tables and supports efficient address aggregation.

Dynamic Host Configuration Protocol (DHCP)

- **Purpose**:
  - Automates IP address assignment and other configuration settings.
  - Allocates temporary IP addresses (lease-based).
- **Process**:
  1. **DHCP Discover**:

        - The client sends a broadcast with a source IP of `0.0.0.0` and destination `255.255.255.255`.
    2. **DHCP Offer**:
        - The server responds with an available IP address, subnet mask, and lease duration.
    3. **DHCP Request**:
        - The client requests the offered IP address.
    4. **DHCP Acknowledgment (ACK)**:
        - Server confirms the IP assignment.
- **Broadcasting**:
    - Uses the broadcast address `255.255.255.255` to communicate with all nodes on the subnet.
- DHCP is a client server protocol . Usually each subnet has a DHCP server , but if not then it has a DHCP relay agent which knows the DHCP assigned to a subnet

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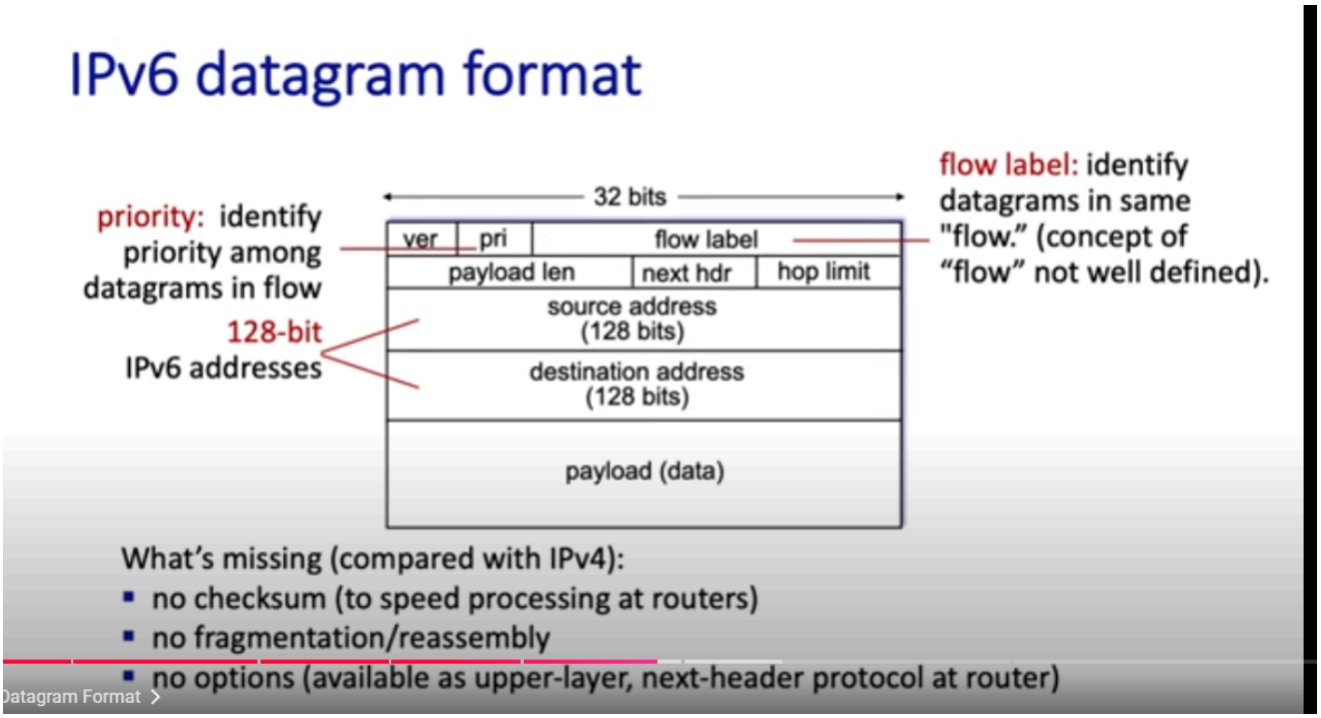
Network Address Translation (NAT)

- **Functionality**:
    - Allows devices in a private network (e.g., `10.0.0.0/24`) to share a single public IP address.
    - a NAT router behaves as a single device to outside world
    - **NAT Translation Table**:
        - Maps private IP and port to public IP and port.
        - Example:
            - Private: `10.0.0.1:3345`.
            - Public: `138.76.29.7:5001`.
    - **How it Works**:
        - Outgoing traffic is translated at the router (private → public).
        - Incoming traffic is mapped back using the NAT table (public → private).
- **Challenges**:
    - Breaks end-to-end connectivity.
    - Affects Peer-to-Peer (P2P) and some real-time applications.
- **Universal Plug and Play (UPnP)**:
    - Allows hosts to dynamically configure NAT mappings for TCP/UDP communication.

IPv6



- **Motivation**:
  - IPv4's 32-bit address space is insufficient for growing Internet demands.
  - **IPv6 Features**:
    - **Expanded Addressing**: 128-bit addresses (unicast, multicast, anycast).
    - **Simplified Header**: Fixed 40-byte header with optional extension headers.
    - No intermediate fragmentation; routers drop oversized packets and notify the sender via ICMPv6.
    - No checksum field, as link-layer and transport-layer checksums suffice.
- **IPv6 Datagram Fields**:
  - **Version**: 4 bits (value = 6).
  - **Traffic Class**: 8 bits for priority or flow labeling.
  - **Flow Label**: 20 bits for flow identification.
  - **Payload Length**: 16 bits for data size.
  - **Next Header**: Identifies the upper-layer protocol (TCP/UDP).
  - **Hop Limit**: Decrements at each hop, similar to IPv4 TTL.
  - **Source/Destination Address**: 128 bits each.
- **Transition from IPv4 to IPv6**:
  - **Dual-stack approach**: Devices support both protocols during the transition.
  - IPv4 headers are not compatible with IPv6. as data would be lost from converting ipv6 to ipv4.
  - Tunneling is used for this problem

How Tunneling Works:

1. **Encapsulation**:
   - Original packet (ipv6) is wrapped inside ipv4 packet (outer packet).

    - Outer packet contains the protocol needed to traverse the intermediate network.
2. **Routing**:
    - Encapsulated packets are routed through the intermediate network.
3. **Decapsulation**:
    - Outer packet is removed at the endpoint, revealing the original packet.

ICMP (Internet Control Message Protocol)

- **Purpose**:
    - Communicates network layer errors and diagnostics.
    - Embedded within IP datagrams.
- **Key Features**:
    - ICMP messages contain:
        - Type and code fields (e.g., echo request/reply, unreachable, TTL expired).
        - IP header and first 8 bytes of the datagram causing the error.
- **Uses**:
    - **Ping**: Echo request/reply messages (Type 8 and Type 0).
    - **Traceroute**: TTL-expired messages (Type 11, Code 0) used to trace packet routes.

Routing Algorithms